



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

25-07-2021

JEE(Main + Advanced) : Nurture Course (PHASE : TEPS, TOPS, I, I(B), II(A) & NURURE (LIVE) I, I(B), II(A))

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	C	B	B	C	C	D	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	D	B	C	A	C	C	A	B	C

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	C	B	B	D	C	C	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	C	D	D	B	D	C	C	C

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	B	D	D	B	D	A	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	C	B	D	C	A	B	D	B

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

As given the block moves with a constant velocity is friction force on block equals to $mg\sin\theta$. As block slides ie friction also equals to μN which is $\mu mg\cos\theta$.

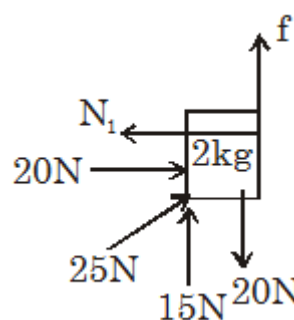
2. Ans (B)

$$a_c = \frac{v^2}{R} = \frac{(2 \times 4)^2}{4} = 16$$

$$a_t = \frac{dv}{dt} = 4$$

$$a_{net} = \sqrt{a_c^2 + a_t^2}$$

3. Ans (D)



$$f + 15 = 20$$

$$f = 5N$$

$$N_1 - 20 = 0$$

$$N_1 = 20$$

4. Ans (C)

$$v_B = \sqrt{2(a_0) \left(\frac{\pi R}{2} \right)} = \sqrt{a_0 \pi R}$$

$$a_B = \sqrt{a_T^2 + a_C^2} = \sqrt{(a_0)^2 + \left(\frac{a_0 \pi R}{R} \right)^2}$$

$$a_B = a_0 \sqrt{1 + \pi^2}$$

5. Ans (B)

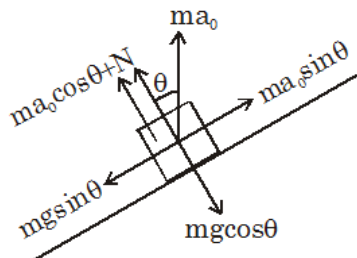
$$v = 1.5 t^2 + 2t$$

$$a_T = 3t + 2$$

$$\alpha = \frac{a_T}{R}$$

7. Ans (C)

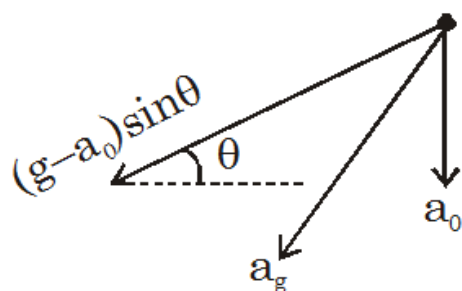
w.r.to lift



$$a_{\text{block/lift}} = (g - a_0) \sin \theta$$

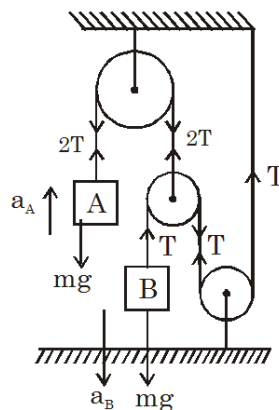
$$t_{AB} = \sqrt{\frac{2L}{(g - a_0) \sin \theta}}$$

w.r.to ground



$$a_g = \sqrt{(g \sin \theta)^2 + (a_0 \cos \theta)^2}$$

8. Ans (C)



$$\varepsilon T \cdot x = 0$$

$$2T \cdot a_A - T \cdot a_B = 0$$

$$a_A = \frac{a_B}{2} \dots (i)$$

$$2T - mg = ma_A$$

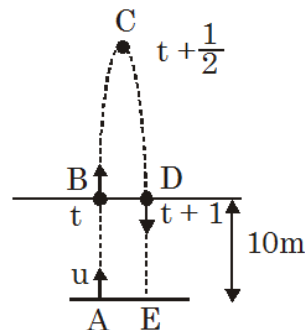
$$2T - mg = \frac{ma_B}{2} \dots (ii)$$

$$mg - T = ma_B \dots (iii)$$

From equation (ii) & (iii)

$$a_B = \frac{2g}{5}$$

9. Ans (D)



$$V_C = 0$$

$$V_B = g \times \frac{1}{2} = 5 \text{ m/s}$$

$$(5)^2 = u^2 - 2 \times g \times 10$$

$$u = 15 \text{ m/s}$$

10. Ans (D)

$$S_t + S_{t+1} = u + \frac{a}{2}(2t-1) + u + \frac{a}{2}(2(t+1)-1)$$

$$100 \text{ cm} = 2u + \frac{a}{2}(2t-1+2t+1)$$

$$100 \text{ cm} = 2u + 2at$$

$$100 \text{ cm} = 2(u+at)$$

$$u + at = 50 \text{ cm/sec}$$

$$V = 50 \text{ cm/sec}$$

11. **Ans (C)**

$$a = \frac{3g - 2g}{5} = 2 \text{ m/s}^2$$

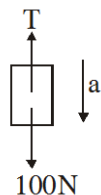
$$25 = 0 + \frac{1}{2} \times 2 \times t^2$$

$$t = 5 \text{ sec}$$

12. **Ans (D)**

$$a = \frac{100}{30} = \frac{10}{3} \text{ m/s}^2$$

From FBD



F.B.D. of C

$$100 - T = 10 \times \frac{10}{3}; T = \frac{200}{3}$$

13. **Ans (B)**

Block is in equilibrium perpendicular to inclined plane

$$N - mg \cos \theta = m(0)$$

acceleration along the inclined plane

$$mg \sin \theta = m(a)$$

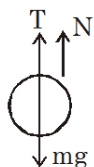
15. **Ans (A)**

$$[y] = \frac{[mv^2]^{1/2}}{[m]^{1/2}}$$

$$[y] = [v] = [LT^{-1}]$$

16. **Ans (C)**

For equilibrium



$$Mg + mg = 6T$$

$$T = \left(\frac{M + m}{6} \right) g$$

For man

$$N \geq 0$$

$$T + N = mg$$

$$N = \left(\frac{5m - M}{6} \right) g$$

$$m \geq M/5$$

17. **Ans (C)**

$$\text{Minimum force required} = \frac{\mu_s mg}{\sqrt{1 + \mu_s^2}}$$

18. **Ans (A)**

Zero as springs are in natural length again.

19. **Ans (B)**



$$\Rightarrow T = 100 \text{ N}$$

20. **Ans (C)**

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = \vec{0}$$

$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = -\vec{F}_4$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (4.00)**

$$v_A \cos 37^\circ = v_B$$

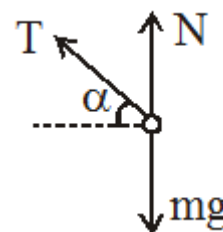
2. **Ans (5.00)**

$$a = g \tan 53^\circ$$

3. **Ans (4.00)**

$$T \cos \alpha = m(\ell \cos \alpha) \omega^2$$

$$N + T \sin \alpha = mg$$



4. **Ans (6.00)**

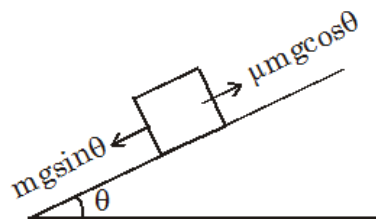
$$\omega = 3\pi^3 = 24\pi$$

$$t = (8)^{\frac{1}{3}} \Rightarrow 2 \text{ sec}$$

$$\int_0^\theta d\theta = \int_0^2 \omega dt \Rightarrow \left[\frac{3\pi t^4}{4} \right]_0^2$$

$$\theta = \frac{3\pi \times 16}{4} \Rightarrow 12\pi \text{ rad.}$$

5. Ans (25.00)



$$v^2 = u^2 + 2as \Rightarrow 0^2 = 10^2 - 2(a)(5)$$

$$\Rightarrow a = 10 \text{ m/s}^2$$

$$a = 10 = \mu g \Rightarrow \mu = 1$$

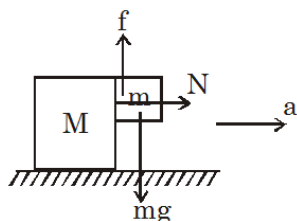
$$mg \sin \theta - \mu mg \cos \theta = ma$$

$$a = -2 \text{ m/s}^2$$

$$v^2 = u^2 + 2as$$

$$0^2 = 10^2 + 2(-2)s \Rightarrow s = 25 \text{ m}$$

6. Ans (25.00)



$$N = ma \dots\dots(1)$$

$$f = mg$$

$$f \leq f_{s \text{ max}}$$

$$mg \leq \mu N$$

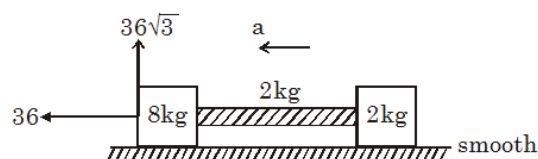
$$mg \leq \mu ma$$

$$a \geq \frac{10}{0.4}$$

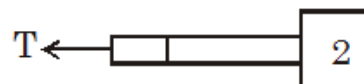
$$a \geq 25$$

$$a_{\min} = \frac{25\text{m}}{\text{s}^2}$$

7. Ans (9.00)



$$a = \frac{36}{12} = 3 \frac{\text{m}}{\text{s}^2}$$



$$T = ma = 3 \times 3 = 9\text{N}$$

8. Ans (8.00)

$$\text{By using } v^2 = u^2 + 2as; 0^2 = 40^2 + 2a(2)$$

$$\text{and } 0^2 = 80^2 + 2a(x) \Rightarrow x = 8\text{m}$$

9. Ans (4.00)



$$50g - T = 50a$$

$$a = 10 - \frac{300}{50} = 4 \text{ m/s}^2$$

10. Ans (3.00)

As no relative motion between blocks is the acceleration of both block will be same



$$\therefore a = \frac{9}{3} = 3\text{m/s}^2$$

$\therefore$ Force of friction between two blocks

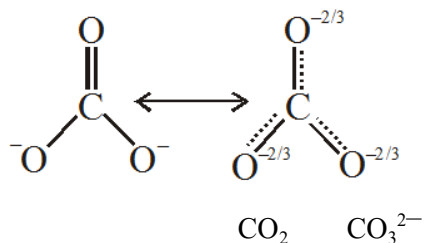
$$= 1 \times 3 = 3\text{N}$$

PART-2 : CHEMISTRY

SECTION-I

9. Ans (C)

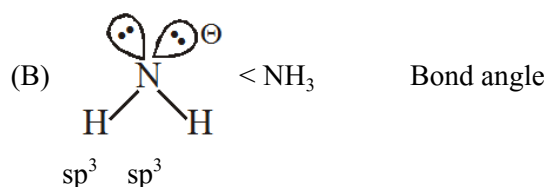
(A) $O=C=O$



Bond angle = 180°

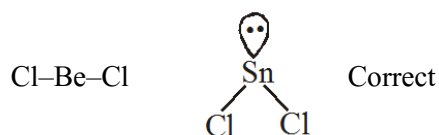
120°

$CO_2 > CO_3^{2-}$



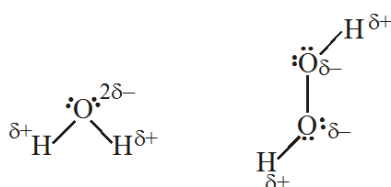
more lone pair on NH_2^- lead to more repulsion and decrease in bond angle.

(C) $BeCl_2$ $SnCl_2$



Since Sn has lone pair bent shape so non zero dipole moment.

(D) H_2O H_2O_2

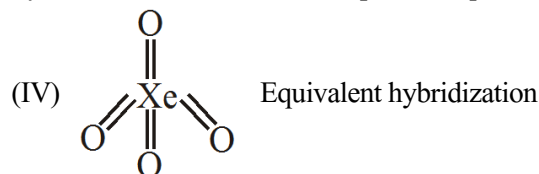
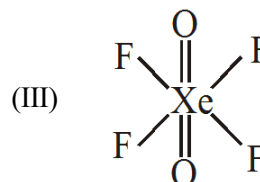
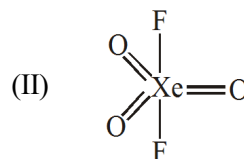
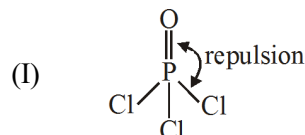


H_2O has more negative charge accumulated on oxygen so more strength of H-Bonding.

$H_2O > H_2O_2$

Option (C) is correct.

11. Ans (A)



Double bond will not affect structure in case of II, III, IV

Option (A) is correct.

12. Ans (A)

$2NH_3 + CO_2 \rightarrow NH_2COONH_4$

$$\frac{n_{NH_3}}{n_{CO_2}} = \frac{2}{1} = \frac{W_{NH_3}}{W_{CO_2}} \times \frac{44}{17} = \frac{2}{1}$$

$$= \frac{W_{NH_3}}{W_{CO_2}} \times \frac{17}{22}$$

13. Ans (C)

Moles of $T_2O = \frac{5.5}{22}$

Moles of neutrons = $\frac{5.5}{22} \times 12 = 3$ mole

No. of neutrons = $3N_A$

14. Ans (D)

Molecular wt. = 240 g/mol

$d = 1.2$ g/mL

Vol of 1 mol molecule = $\frac{240}{1.2}$ mL

Vol of 1 (N_A) molecule

Vol of $2N_A$ molecule

$$= \frac{240}{1.2 N_A} \times 2N_A \text{ mL} = 400 \text{ mL} = 0.4 \text{ L}$$

16. **Ans (B)**

$$\frac{n_{H_2}}{3} < \frac{n_{N_2}}{1} \left| \frac{n_{N_2}}{n_{H_2}} > \frac{1}{3} \right.$$

$$\frac{W_{N_2}}{W_{H_2}} > \frac{14}{3}$$

17. **Ans (D)**

$$\begin{array}{ccc} O_3 & O_2 & \\ \text{Mole ratio :} & 1 & 2 \end{array}$$

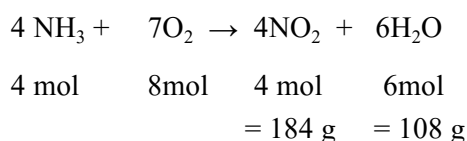
$$[VD]_{\text{mix}} = \frac{M_{\text{avg}}}{2} = \frac{48 + 64}{2 \times 3} = \frac{112}{2 \times 3} = \frac{56}{3}$$

18. **Ans (C)**

$$\begin{array}{ccc} 2H_2 & + & O_2 \rightarrow 2H_2O \\ \text{mol} & 2 & 1 \\ \text{mass} & 2 \times 2 & 1 \times 32 \text{ g} \\ & 10 \text{ g} & 80 \text{ g} \\ \text{mol} & 5 & 2.5 \text{ mol} \end{array}$$

$$5 \times \frac{80}{100} \times 18 = 72 \text{ gm}$$

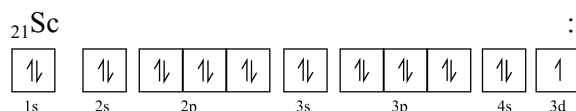
19. **Ans (C)**



PART-2 : CHEMISTRY

SECTION-II

2. **Ans (11.00)**



3. **Ans (72.00)**

$$XM^2 \Rightarrow m = \frac{n+2}{2} \Rightarrow \frac{10+2}{2} = 6$$

$$2 \times (6)^2 = 72$$

5. **Ans (7.00)**

$$\Delta EN = |2.5 - 2.1|$$

$$= 0.4$$

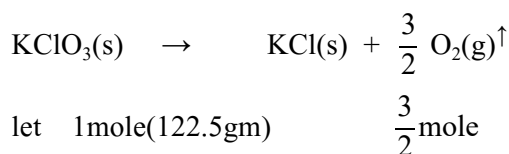
$$\%IC = 16 |\Delta EN| + 3.5 |\Delta EN|^2$$

$$= 16 \times 0.4 + 3.5 \times 0.4^2$$

$$= 6.96\%$$

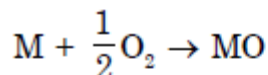
$$\approx 7\%$$

7. **Ans (39.18)**



$$\% \text{ loss} = \frac{\frac{3}{2} \times 32}{122.5} \times 100 = 39.18$$

8. **Ans (24.00)**



$$\text{Mole of } O_2 \text{ used} = \frac{8}{32} = 0.25 \text{ mole}$$

$$\text{Moles of metal reacted} = 2 \times n_{O_2}$$

$$= 2 \times 0.25 = 0.5 \text{ mole}$$

$$\therefore \text{At. weight of metal} = \frac{12}{0.5} = 24$$

9. **Ans (20.00)**

$$\% \text{ of C} = \frac{1}{5} \times 100 = 20\%$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**

$$9^{\log_3(\log_2 x)} = \log_2 x - (\log_2 x)^2 + 1$$

$$(\log_2 x)^2 = (\log_2 x) - (\log_2 x)^2 + 1$$

$$\log_2 x = t$$

$$2t^2 - t - 1 = 0, (2t+1)(t-1) = 0$$

$$t = 1 \text{ or } t = -\frac{1}{2} \text{ (rejected)}$$

$$\Rightarrow \log_2 x = 1 \Rightarrow x = 2$$

2. **Ans (A)**

$$\log_3(xyz) = -1 \Rightarrow xyz = \frac{1}{3}$$

$$\text{Now } (xyz)^{x+y+z} = 1 \Rightarrow x+y+z = 0$$

$$\Rightarrow x^3 + y^3 + z^3 = 3xyz$$

$$\Rightarrow x^3 + y^3 + z^3 = 1 \Rightarrow \log_{xyz}(x^3 + y^3 + z^3) = 0$$

3. **Ans (A)**

$$a_0 = \cos \theta, a_1 = \sqrt{\frac{1}{2}(1 + \cos \theta)} = \cos \frac{\theta}{2}$$

$$a_2 = \sqrt{\frac{1}{2}\left(1 + \cos \frac{\theta}{2}\right)} = \cos \frac{\theta}{2^2}$$

$$\vdots$$

$$a_n = \frac{\cos \theta}{2^n}$$

$$\Rightarrow \frac{\sqrt{1 - a_0^2}}{(a_1 a_2 \dots a_n) \times \sqrt{1 - a_n^2}}$$

$$= \frac{\sqrt{1 - \cos^2 \theta}}{\left(\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n}\right) \sqrt{1 - \cos^2 \frac{\theta}{2^n}}}$$

$$= \frac{\sin \theta}{\left(\frac{\sin\left(2^n \cdot \frac{\theta}{2^n}\right)}{2^n \sin \frac{\theta}{2^n}}\right) \sin\left(\frac{\theta}{2^n}\right)} = 2^n$$

4. **Ans (B)**

$$-3 \leq 3\sin x - 4\cos x + 2 \leq 7$$

$$\Rightarrow \frac{1}{3\sin x - 4\cos x + 2} \in \left(-\infty, -\frac{1}{3}\right] \cup \left[\frac{1}{7}, +\infty\right)$$

5. **Ans (D)**

Since $x^2 + 5x + 2 = 0$ has irrational roots.

$\therefore$ Both equations will have both roots common.

$$\Rightarrow \frac{a}{1} = \frac{(b+2)}{5} = \frac{c}{2}$$

$$a_{\min} = 1$$

$$b_{\min} = 3$$

$$c_{\min} = 2$$

$$\Rightarrow (a + b + c)_{\min} = 6$$

6. **Ans (D)**

when $x = \text{odd}$,

$$\Rightarrow \text{odd} - \text{even} - \text{even} + \text{odd} + \text{odd} = 0$$

(not possible)

when $x = \text{even}$

$$\Rightarrow \text{even} - \text{even} - \text{even} + \text{even} + \text{odd} = 0$$

(not possible)

$\Rightarrow$ no integral solution

7. **Ans (B)**

$$3(1 - \sin^2 x) = 8\sin x$$

$$\text{Let } \sin x = S$$

$$3(1 - S^2) = 8S$$

$$3S^2 + 8S - 3 = 0$$

$$3S^2 + 9S - S - 3 = 0$$

$$3S(S + 3) - (S + 3) = 0$$

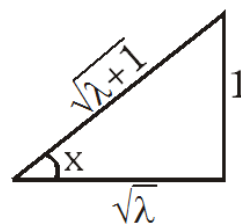
$$\sin x = -3, \text{ (rejected)}$$

$$\Rightarrow \sin x = \frac{1}{3} \Rightarrow \text{two solutions}$$

8. **Ans (D)**

$$\lambda \tan^2 x + \frac{1}{\lambda \tan^2 x} = 2$$

$$\Rightarrow \lambda \tan^2 x = 1 \Rightarrow \tan x = \frac{1}{\sqrt{\lambda}}$$



$$\sin x = \frac{1}{\sqrt{\lambda + 1}} \quad \& \quad \cos x = \frac{\sqrt{\lambda}}{\sqrt{\lambda + 1}}$$

$$\sin^2 x + 5 \cos^2 x + 3 \sin x \cos x$$

$$= \frac{1}{(\lambda + 1)} + \frac{5\lambda}{(\lambda + 1)} + \frac{3\sqrt{\lambda}}{(\lambda + 1)}$$

$$= \frac{1 + 5\lambda + 3\sqrt{\lambda}}{(\lambda + 1)}$$

9. **Ans (A)**

Given A.P. is $a_1, a_2, a_3, \dots, a_{15}$

$$a_1 + a_{15} = a_2 + a_{14} = \dots = 2a_8$$

$$a_1 + a_{15} + a_8 = 15$$

$$\Rightarrow 3a_8 = 15 \Rightarrow a_8 = 5$$

$$\Rightarrow a_1 + a_{15} = 10$$

$$\text{Consider } a_2 + a_3 + a_8 + a_{13} + a_{14}$$

$$= 2a_8 + 2a_8 + a_8 = 25$$

10. **Ans (D)**

Simplify the equation to the form $xy + 42 = 9y$,

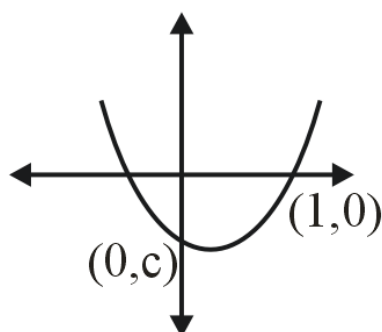
$$\text{then } y = \frac{42}{9-x}$$

y is a positive integer implies that $9-x$ is a positive divisor of 42, so

$$9-x = 1, 2, 3, 6, 7 \quad \text{i.e. } x = 8, 7, 6, 3, 2$$

Correspondingly, $y = 42, 21, 14, 7, 6$. By checking, the five solutions satisfy the original equation, so the answer is 5.

11. **Ans (B)**



$$a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow a + b + c = 0$$

$$\Rightarrow f(1) = 0$$

and c must be negative.

$$\Rightarrow f(0) < 0$$

$\Rightarrow$ one root is positive and other is negative.

12. **Ans (B)**

$$9 \sec^2 \theta + 4 \operatorname{cosec}^2 \theta$$

$$13 + 9 \tan^2 \theta + 4 \cot^2 \theta$$

$$\text{min value} = 13 + 2\sqrt{9 \times 4} = 25$$

13. **Ans (C)**

$$\sin x = 1 - \sin^2 x$$

$$\Rightarrow \sin x = \cos^2 x$$

$$\text{Now } \cos^2 x + \cos^4 x$$

$$= \sin x + \sin^2 x$$

$$= 1$$

14. **Ans (B)**

$$y = 7 \cos^2 x + 6 \sin x \cos x - \sin^2 x$$

$$= 3 \sin 2x + 4 \cos 2x + 3$$

The maximum value of y is 8

$$\text{Also } \log_{\sqrt{2}} 8 = \log_{\sqrt{2}} (\sqrt{2})^6 = 6$$

15. **Ans (D)**

$$2 \sin 54^\circ \cdot \cos 7^\circ - 2 \sin 18^\circ \cos 7^\circ$$

$$= 2 \cos 7^\circ [\sin 54^\circ - \sin 18^\circ]$$

$$= 2 \cos 7^\circ [2 \cos 36^\circ \cdot \sin 18^\circ]$$

$$= 4 \cos 7^\circ \left(\frac{\sqrt{5}+1}{4} \right) \left(\frac{\sqrt{5}-1}{4} \right)$$

$$= \cos 7^\circ$$

16. **Ans (C)**

$$\left(\frac{1}{\sqrt{2}} \right)^{\tan x} = (2)^{\sin x}$$

$$\Rightarrow (2)^{\frac{-\tan x}{2}} = 2^{\sin x}$$

$$\Rightarrow \frac{-\tan x}{2} = \sin x$$

$$\Rightarrow \sin x = 0 \text{ or } \cos x = \frac{-1}{2}$$

$$\Rightarrow x = 0, \pi, \frac{2\pi}{3}$$

17. **Ans (A)**

$$\sec 80^\circ - \sqrt{3} \cos 80^\circ = \frac{1}{\cos 80^\circ} - \frac{\sin 60^\circ}{\cos 60^\circ} \frac{1}{\sin 80^\circ}$$

$$= \frac{\sin 20^\circ}{\cos 60^\circ \cos 80^\circ \sin 80^\circ} = 4$$

18. **Ans (B)**

$$\text{Given } \frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{5n+3}{7n+1}$$

$$\Rightarrow \frac{2a_1 + (n-1)d_1}{2a_2 + (n-1)d_2} = \frac{5n+3}{7n+1}$$

$$\Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d_1}{a_2 + \left(\frac{n-1}{2}\right)d_2} = \frac{5n+3}{7n+1} \quad \dots(i)$$

To get ratio of 7th term, we must have $\frac{n-1}{2} = 6$

$$\text{i.e., } n = 13$$

$$\Rightarrow \frac{a_1 + 6d_1}{a_2 + 6d_2} = \frac{5(13)+3}{7(13)+1} = \frac{68}{92} = \frac{17}{23}$$

19. Ans (D)

$\therefore$ first term $a_1 = 2$ {given}

$$\therefore a_n = \frac{a_{n-1}^2}{a_{n-2}} \Rightarrow \frac{a_n}{a_{n-1}} = \frac{a_{n-1}}{a_{n-2}}$$

ratio of consecutive terms is constant

$\therefore$ sequence is G.P.

Let $a_2 = x$

then common ratio $= \frac{x}{2}$

$$\therefore a_5 = 2 \left(\frac{x}{2} \right)^4 \Rightarrow a_5 = \frac{x^4}{8}$$

$\therefore a_5 \rightarrow +ve$ integer

$\therefore x$ must be an even & integer.

$\therefore$ Given $\Rightarrow a_5 \leq 512$

$$\frac{x^4}{8} \leq 2009 \Rightarrow x^4 \leq 4096$$

$$\Rightarrow x \leq 8 \quad \{ \because x \in \text{even} \}$$

$\therefore x = 2, 4, 6, 8$

So $a_5 = 2, 32, 162, 512$

20. Ans (B)

$$\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + \frac{8}{\tan 24^\circ}$$

$$\tan 3^\circ + 2 \tan 6^\circ + 4 \tan 12^\circ + \frac{8(1 - \tan^2 12^\circ)}{2 \tan 12^\circ}$$

$$\tan 3^\circ + 2 \tan 6^\circ + \frac{4}{\tan 12^\circ}$$

$$= \tan 3^\circ + 2 \tan 6^\circ + \frac{2(1 - \tan^2 6^\circ)}{\tan 6^\circ} = \tan 3^\circ + \frac{2}{\tan 6^\circ}$$

$$= \cot 3^\circ \Rightarrow \theta = 3$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5.00)

$$\alpha^2 + \alpha - 3 = 0 \Rightarrow \alpha^2 = 3 - \alpha$$

$$\beta^2 + \beta - 3 = 0 \Rightarrow \beta^2 = 3 - \beta$$

$$\alpha^3 - 4\beta^2 + 24 = \alpha(3 - \alpha) - 4(3 - \beta) + 24$$

$$= 3\alpha + \alpha - 3 - 12 + 4\beta + 24$$

$$= -4 - 3 - 12 + 24 = 5$$

2. Ans (20.00)

$$\frac{t_3}{t_6} = \frac{1}{8}$$

$$\frac{ar^2}{ar^5} = \frac{1}{8}$$

$$r^3 = 8$$

$$r = 2$$

$$t_1 + t_5 = 170$$

$$a + ar^4 = 170$$

$$a + 16a = 170$$

$$17a = 170$$

$$a = 10$$

$$t_2 = ar$$

$$= 10 \times 2$$

$$t_2 = 20$$

3. Ans (6.00)

$$\sin \alpha = 1 \quad \sin \beta = 1$$

$$\cos \alpha = 0 \quad \cos \beta = 0$$

4. Ans (1.25)

$$\sin^2 x + 1 - \cos^2 \left(x + \frac{\pi}{3} \right) + \cos x \cdot \left(x + \frac{\pi}{3} \right)$$

$$\left(x + \frac{\pi}{3} \right)$$

$$= 1 - \cos \left(2x + \frac{\pi}{3} \right) \cos \left(\frac{\pi}{3} \right)$$

$$1 - \cos \left(2x + \frac{\pi}{3} \right) \cos \left(\frac{\pi}{3} \right)$$

$$+ \frac{1}{2} \left\{ \cos \left(2x + \frac{\pi}{3} \right) + \cos \left(\frac{\pi}{3} \right) \right\}$$

$$+ \frac{1}{2} \left\{ \cos \left(2x + \frac{\pi}{3} \right) + \cos \left(\frac{\pi}{3} \right) \right\}$$

$$= 1 - \frac{1}{2} \cos \left(2x + \frac{\pi}{3} \right) + \frac{1}{4} + \frac{1}{2} \cos \left(2x + \frac{\pi}{3} \right)$$

$$= 1 - \frac{1}{2} \cos \left(2x + \frac{\pi}{3} \right) + \frac{1}{4} + \frac{1}{2} \cos \left(2x + \frac{\pi}{3} \right)$$

$$= \frac{5}{4} = \frac{5}{4}$$

5. **Ans (2.00)**

$$2\sin^2\theta = 1 + \sin\theta \Rightarrow \sin\theta = 1 \text{ or } \sin\theta = -\frac{1}{2} - \frac{1}{2}$$

$\sin\theta = 1$ is not acceptable because its a non constant AP

$$\text{hence } \sin\theta = -\frac{1}{2} - \frac{1}{2} \Rightarrow 2 \text{ values of '}\theta\text{'}$$

6. **Ans (1.50)**

$$\text{Given } a_1 = 2, a_{10} = 3 \Rightarrow d = \frac{1}{9}d = \frac{1}{9}$$

$$\text{So } a_3 = a_1 + 2d = \frac{20}{9} \frac{20}{9}$$

$$\text{Given } h_1 = 2, h_{10} = 3$$

$$\text{So corresponding A.P. } T_1 = \frac{1}{2} \frac{1}{2}, T_{10} = \frac{1}{3} \frac{1}{3}$$

$$\Rightarrow D = -\frac{1}{54} D = -\frac{1}{54} \text{ So } T_8 = T_1 + 7D = \frac{10}{27} \frac{10}{27}$$

$$\therefore h_8 = \frac{27}{10} h_8 = \frac{27}{10}$$

$$\text{So } a_3 h_8 = \frac{20}{9} \times \frac{27}{10} = 6 \frac{20}{9} \times \frac{27}{10} = 6$$

$$\frac{a_3 h_8}{4} = \frac{6}{4} = 1.50 \frac{a_3 h_8}{4} = \frac{6}{4} = 1.50$$

7. **Ans (1.00)**

$$y = \sqrt{\sin^4 x + 4(1 - \sin^2 x)} - \sqrt{\cos^4 x + 4(1 - \cos^2 x)}$$

$$y = \sqrt{\sin^4 x + 4(1 - \sin^2 x)} - \sqrt{\cos^4 x + 4(1 - \cos^2 x)}$$

$$= \sqrt{(2 - \sin^2 x)^2} - \sqrt{(2 - \cos^2 x)^2}$$

$$= \sqrt{(2 - \sin^2 x)^2} - \sqrt{(2 - \cos^2 x)^2}$$

$$= (2 - \sin^2 x) - (2 - \cos^2 x)$$

$$y = \cos^2 x - \sin^2 x = \cos 2x$$

$$y_{\max} = 1$$

8. **Ans (3.00)**

$$\frac{\sin 2\alpha}{\sin 2\beta} = 4 \frac{\sin 2\alpha}{\sin 2\beta} = 4$$

$$\frac{\sin 2\alpha + \sin 2\beta}{\sin 2\alpha - \sin 2\beta} = \frac{5 \sin 2\alpha + \sin 2\beta}{3 \sin 2\alpha - \sin 2\beta} = \frac{5}{3}$$

(using C and D)

$$\frac{\frac{1}{2} \sin(\alpha + \beta) \cos(\alpha - \beta)}{\frac{1}{2} \cos(\alpha + \beta) \sin(\alpha - \beta)} = \frac{5}{3}$$

$$\frac{\frac{1}{2} \sin(\alpha + \beta) \cos(\alpha - \beta)}{\frac{1}{2} \cos(\alpha + \beta) \sin(\alpha - \beta)} = \frac{5}{3}$$

$$\frac{\tan(\alpha + \beta)}{\tan(\alpha - \beta)} = \frac{5 \tan(\alpha + \beta)}{3 \tan(\alpha - \beta)} = \frac{5}{3}$$

9. **Ans (2.00)**

$$x^2 + ax + a + 1 = 0$$

$$D = a^2 - 4(1)(a + 1)$$

$$\Rightarrow D = a^2 - 4a - 4$$

$$\Rightarrow D = a^2 - 4a + 4 - 8$$

$$\Rightarrow D = (a - 2)^2 - 8$$

For integral roots and $a \in \mathbb{I}$

Discriminant must be perfect square

$$\therefore 8 = (a - 2)^2 - D$$

Possible when

$$(a - 2)^2 = 9 \text{ and } D = 1$$

$$a - 2 = \pm 3$$

$$a = 5, -1$$

10. **Ans (0.00)**

a, b, c are +ve real nos.

$$\underline{ax^2} + \underline{b|x|} + \underline{c} = 0$$

$$\underline{+ve} \quad \underline{+ve} \quad \underline{+ve}$$

$\therefore$ LHS is always positive (i.e. > 0)

$\therefore$ No real solution possible.

$\therefore$ Number of real roots = 0